KAIZEN MATHS

Lesson starter

Transposition of Formulae: Changing the Subject

Foundations

Name _____

S1

Make the named variable the subject of each formula.

Make x the subject.

$y = x + 6$

S2

Make the named variable the subject of each formula.

Make x the subject.

$y = 6x$

S3

Make the named variable the subject of each formula.

Make x the subject.

$y = 2x + 4$

S4

Make the named variable the subject of each formula.

Make x the subject.

$y = x + 9$

Show your working

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ALG-0852-L1 · S

Show your working

KAIZEN MATHS

Exit ticket

Transposition of Formulae: Changing the Subject

Foundations

Name _____

EA1

Make the named variable the subject of each formula.

Make x the subject.

$y = 7x$

EA2

Make the named variable the subject of each formula.

Make x the subject.

$y = 7x - 7$

Show your working

KAIZEN MATHS

Exit ticket

Transposition of Formulae: Changing the Subject

Foundations

Name _____

EB1

Make the named variable the subject of each formula.

Make x the subject.

$y = x - 9$

Show your working

KAIZEN MATHS

Exit ticket

Transposition of Formulae: Changing the Subject

Foundations

Name _____

EA1

Make the named variable the subject of each formula.

Make x the subject.

$y = 7x$

EA2

Make the named variable the subject of each formula.

Make x the subject.

$y = 7x - 7$

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Show your working

KAIZEN MATHS

Exit ticket

Transposition of Formulae: Changing the Subject

Foundations

Name _____

EB1

Make the named variable the subject of each formula.

Make x the subject.

$y = x - 9$

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Show your working

Teacher answers & working

Transposition of Formulae: Changing the Subject · Foundations

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S1

Make the named variable the subject of each formula.

Make x the subject.

$y = x + 6$

Answer:

$x = y - 6$

Working:

Start with the formula and identify the term containing x.

$y = x + 6$

Undo the addition by choosing the inverse operation: subtract 6 on both sides.

$y - 6 = x + 6 - 6$

Simplify the side containing x.

$y - 6 = x$

Write the formula with the new subject first.

$x = y - 6$

S2

Make the named variable the subject of each formula.

Make x the subject.

$y = 6x$

Answer:

$x = \frac{y}{6}$

Working:

Start with the formula and notice that x is multiplied by 6.

$y = 6x$

Undo multiplication by 6 by dividing both sides by 6.

$\frac{y}{6} = \frac{6x}{6}$

The 6 in the numerator and denominator cancel on the right-hand side.

$\frac{y}{6} = x$

Write the formula with x as the subject.

$x = \frac{y}{6}$

Teacher answers & working

Transposition of Formulae: Changing the Subject · Foundations

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S3

Make the named variable the subject of each formula.

Make x the subject.

$y = 2x + 4$

Answer:

$x = \frac{y-4}{2}$

Working:

Start with the formula. The subject x is multiplied by 2 and then 4 is added.

$y = 2x + 4$

Undo the addition first.

$y - 4 = 2x$

Now undo multiplication by 2 by dividing both sides by 2.

$\frac{y-4}{2} = \frac{2x}{2}$

Simplify the right-hand side.

$\frac{y-4}{2} = x$

Write x as the subject.

$x = \frac{y-4}{2}$

S4

Make the named variable the subject of each formula.

Make x the subject.

$y = x + 9$

Answer:

$x = y - 9$

Working:

Start with the formula and identify the term containing x.

$y = x + 9$

Undo the addition by choosing the inverse operation: subtract 9 on both sides.

$y - 9 = x + 9 - 9$

Simplify the side containing x.

$y - 9 = x$

Write the formula with the new subject first.

$x = y - 9$

Teacher answers & working

Transposition of Formulae: Changing the Subject · Foundations

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EA1

Make the named variable the subject of each formula.

Make x the subject.

$y = 7x$

Answer:

$x = \frac{y}{7}$

Working:

Start with the formula and notice that x is multiplied by 7.

$y = 7x$

Undo multiplication by 7 by dividing both sides by 7.

$\frac{y}{7} = \frac{7x}{7}$

The 7 in the numerator and denominator cancel on the right-hand side.

$\frac{y}{7} = x$

Write the formula with x as the subject.

$x = \frac{y}{7}$

EA2

Make the named variable the subject of each formula.

Make x the subject.

$y = 7x - 7$

Answer:

$x = \frac{y+7}{7}$

Working:

Start with the formula. The subject x is multiplied by 7 and then 7 is subtracted.

$y = 7x - 7$

Undo the subtraction first.

$y + 7 = 7x$

Now undo multiplication by 7 by dividing both sides by 7.

$\frac{y+7}{7} = \frac{7x}{7}$

Simplify the right-hand side.

$\frac{y+7}{7} = x$

Write x as the subject.

$x = \frac{y+7}{7}$

Teacher answers & working

Transposition of Formulae: Changing the Subject · Foundations

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EB1

Make the named variable the subject of each formula.

Make x the subject.

$y = x - 9$

Answer:

$x = y + 9$

Working:

Start with the formula and identify the term containing x.

$y = x - 9$

Undo the subtraction by choosing the inverse operation: add 9 on both sides.

$y + 9 = x - 9 + 9$

Simplify the side containing x.

$y + 9 = x$

Write the formula with the new subject first.

$x = y + 9$